

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Annexure A

16
20
Q. J. H. Total Marks: 25

Comparative Analysis

Code of Conduct:

I T. Karimulla / 192065046 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Karimulla

① Star and bus are two commonly studied network topologies. In a star topology, all devices are connected to a central switch or hub, whereas in a bus topology, all devices share a common backbone cable. They differ considerably in cost, reliability, and network performance.

- Star topology requires a central networking device.
- Bus topology uses a single backbone cable.
- Star provides better performance because data transmission is centrally managed.
- Bus topology is economical for small networks.
- Star is generally more suitable where reliability and performance are important.

Conclusion:

Although bus topology is cheaper and simpler, star topology provides better reliability, performance and fault isolation.

② Mesh and star topologies differ mainly the way devices are interconnected and how they handle failures. Mesh topology provides multiple communication paths between devices, while star topology depends on a central switch or hub.

- Mesh topology provides multiple paths for data transmission
- Failure of one mesh link does not necessarily interrupt communication
- In star topology, failure of one node normally does not affect other nodes
- Failure of the central switch can stop the star network
- Mesh is more appropriate for critical and highly reliable environments

Conclusion:

Mesh topology offers better fault tolerance than star topology, but its complex structure makes it more difficult to implement and maintain.

Bus and Mesh topologies have significant differences in installation complexity. Bus topology has a simple structure based on a shared backbone, whereas mesh topology contains numerous connections between devices.

- Bus topology requires minimal connections.
- Mesh topology requires many cables and ports.
- Mesh installation becomes increasingly difficult as devices are added.
- Bus topology is easier and faster to install.
- Mesh topology requires greater planning and resources.

Conclusion:

Bus topology is easier and less expensive to install while mesh topology provides greater connectivity at the cost of increased complexity.

② Mesh and star .1

④ Scalability is an important factor when a network that may need future expansion. star topology provides greater flexibility, while star topology mainly depends on the capacity of its central networking device.

- Hybrid topology can combine star, bus, mesh or other structures
- New network sections can be added according to requirements
- Star topology depends on the available ports of the central switch
- Additional hardware may be required when the central device reaches its limit.
- Hybrid topology provides greater flexibility for organizational networks

Conclusion:

Hybrid topology is generally more scalable because it can combine different topologies and expand according to network requirements